Engineered Features Reference

Urban Rainwater Harvesting Estimation, Bengaluru: rainfall ML features

These nine features were proposed by the Antigravity agent for the project's two rainfall ML tasks: the **temporal scenario model** (probabilistic quantiles for dry, normal and wet years, seasonal totals and dry-spell durations) and the **spatial adjustment model** (adjusting regional rainfall to a building's location). Every feature uses only information from day t-1 or earlier, so that it does not leak the value being predicted. Some features expand into several columns, giving about 14 columns in total.

Summary

#	Feature	Group	Columns
1	Antecedent Dry Weather Period (ADWP)	Group 1	1
2	Multi-Horizon Rolling Cumulative Precipitation	Group 1	4 (k = 3, 7, 14, 30)
3	Rolling Volatility and Coefficient of Variation	Group 1	1
4	Harmonic Day-of-Year Encodings	Group 2	2 (sin, cos)
5	Season Progression Index	Group 2	1
6	East-West and North-South Spatial Gradient Indices	Group 3	2 (E-W, N-S)
7	Spatial Concurrence (Coherence) Ratio	Group 3	1
8	Higher-Order Markov Transition State	Group 4	1 (categorical, 4 states)
9	Cumulative Season Departure Z-Score	Group 4	1

Notation

P_t	Daily rainfall (mm) on day t.
tau	Dry-day threshold, 0.1 mm. A day is dry if P is below tau.
epsilon	Small positive constant that avoids division by zero.
DOY_t	Day of year for day t.
Central cell	IMD grid cell at 13.00 N, 77.50 E (column rain_central_bbmp_mm).
South cell	IMD grid cell at 12.75 N, 77.50 E (column rain_south_ecity_mm).
East cell	IMD grid cell at 13.00 N, 77.75 E (column rain_east_whitefield_mm).

Feature Definitions

1. Antecedent Dry Weather Period (ADWP)

Group 1: Antecedent rainfall memory and persistence

Definition	Number of consecutive days, ending the day before day t, on which rainfall stayed below the dry-day threshold of 0.1 mm. The count resets to zero after any day at or above the threshold.
Formula	$\mathrm{ADWP}_t = 0 \quad \text{if } P_{t-1} \geq \tau$ $\mathrm{ADWP}_t = \mathrm{ADWP}_{t-1} + 1 \quad \text{if } P_{t-1} < \tau$ P is daily rainfall in mm; tau = 0.1 mm is the dry-day threshold.
Purpose	Tracks how long the roof catchment has been dry before day t.
Why it helps	**Hydrology:** longer dry periods mean more dust on roofs, so more first-flush rejection and lower initial runoff efficiency. **ML:** directly models consecutive dry days and the tail of dry-spell lengths (Pareto or exponential behaviour).

2. Multi-Horizon Rolling Cumulative Precipitation

Group 1: Antecedent rainfall memory and persistence

Definition	Total rainfall over the k days before day t, computed for k = 3, 7, 14 and 30 days (four features).
Formula	$P_{\mathrm{roll},k}(t) = \sum_{i=1}^{k} P_{t-i}, \qquad k \in \{3, 7, 14, 30\}$ The sum starts at i = 1, so day t itself is never included.
Purpose	Quantifies moisture accumulation and atmospheric persistence from short (synoptic waves) to medium (intra-seasonal oscillations) horizons.
Why it helps	Rain in Bengaluru often arrives in clusters driven by low-pressure troughs. High 7-day and 14-day totals signal an active monsoon spell, while low totals signal a break spell.

3. Rolling Volatility and Coefficient of Variation

Group 1: Antecedent rainfall memory and persistence

Definition	Ratio of the standard deviation to the mean of daily rainfall over the previous 14 days. A small constant epsilon avoids division by zero.
Formula	$CV_k(t) = \frac{\sigma_k(P_{t-1}, \ldots, P_{t-k})}{\mu_k(P_{t-1}, \ldots, P_{t-k}) + \varepsilon}, \qquad k = 14$ sigma_k and mu_k are the standard deviation and mean over the same k-day window.
Purpose	Measures rainfall intermittency and volatility.
Why it helps	Separates continuous, gentle monsoon rain (CV about 1) from erratic, violent convective pre-monsoon thunderstorms (CV well above 2), which can make rooftop gutters and tanks overflow almost instantly.

4. Harmonic Day-of-Year Encodings

Group 2: Cyclical calendar and monsoon phenology

Definition	Represents the day of the year as a point on a circle using a sine and a cosine with a period of 365.25 days (two features).
Formula	$$\sin_{\text{doy}}(t) = \sin\left(\frac{2\pi \cdot DOY_t}{365.25}\right)$$ $$\cos_{\text{doy}}(t) = \cos\left(\frac{2\pi \cdot DOY_t}{365.25}\right)$$ DOY_t is the day of year (1 to 366).
Purpose	Maps the annual cycle onto continuous circular coordinates.
Why it helps	Removes the artificial jump between 31 December (DOY 365) and 1 January (DOY 1), so tree-based and gradient boosted models can learn smooth annual seasonality.

5. Season Progression Index

Group 2: Cyclical calendar and monsoon phenology

Definition	Fraction of the way through the current season, from 0 at the season start to 1 at the season end. Seasons used: Winter (Jan-Feb), Pre-Monsoon (Mar-May), SW Monsoon (Jun-Sep), NE Post-Monsoon (Oct-Dec).
Formula	$$\text{Prog}_{\text{season}}(t) = \frac{DOY_t - DOY_{\text{start}}}{DOY_{\text{end}} - DOY_{\text{start}}} \in [0, 1]$$ DOY_start and DOY_end are the first and last day of the season that contains day t.
Purpose	Measures relative progress through Bengaluru's specific seasons.
Why it helps	Captures the onset, peak and withdrawal stages of the monsoons. For example, September-October is historically Bengaluru's wettest peak.

6. East-West and North-South Spatial Gradient Indices

Group 3: Spatial disparity and micro-zone gradients

Definition	Difference in previous-day rainfall between the east and central cells (east-west), and between the central and south cells (north-south).
Formula	$$\Delta_{\text{EW}}(t-1) = P_{\text{east},\,t-1} - P_{\text{central},\,t-1}$$ $$\Delta_{\text{NS}}(t-1) = P_{\text{central},\,t-1} - P_{\text{south},\,t-1}$$ Delta EW uses the east minus central cell; Delta NS uses the central minus south cell. Cell locations are listed in the notation section.
Purpose	Quantifies localized micro-climate asymmetry across Bengaluru.
Why it helps	East Bengaluru (Whitefield and KR Puram) often records higher storm totals during localized convective events than western or southern zones. This directly assists the spatial adjustment model.

7. Spatial Concurrence (Coherence) Ratio

Group 3: Spatial disparity and micro-zone gradients

Definition	Smallest divided by largest of the previous day's rainfall across the three cells. Close to 1 means the rain was similar everywhere; close to 0 means it was concentrated in one place.
Formula	$$\rho_{\text{spatial}}(t-1) = \frac{\min(P_{\text{central}},\, P_{\text{south}},\, P_{\text{east}})_{t-1}}{\max(P_{\text{central}},\, P_{\text{south}},\, P_{\text{east}})_{t-1} + \varepsilon}$$ epsilon is a small constant that avoids division by zero on completely dry days.
Purpose	Measures the spatial uniformity of the previous day's rainfall event.
Why it helps	Distinguishes city-wide synoptic depressions (rho close to 1) from hyper-localized convective thunderstorms (rho close to 0).

8. Higher-Order Markov Transition State

Group 4: Two-stage precipitation and departure features

Definition	A wet/dry indicator for each of the previous two days (wet if rainfall is at or above the threshold), combined into one of four states: (0,0), (0,1), (1,0), (1,1).
Formula	$$W_{t-i} = \mathbf{1}(P_{t-i} \geq \tau) \in \{0, 1\}$$ $$\text{State}_{\text{Markov}}(t) = (W_{t-1},\, W_{t-2}) \in \{(0,0),\, (0,1),\, (1,0),\, (1,1)\}$$ 1(.) is the indicator function: 1 when the condition holds, otherwise 0. This is a categorical feature.
Purpose	Encodes the two-day rainfall state history.
Why it helps	Standard meteorological practice uses a two-stage model: first predict whether rain occurs given the last two days, then predict the amount using a Gamma or Tweedie distribution.

9. Cumulative Season Departure Z-Score

Group 4: Two-stage precipitation and departure features

Definition	Standardized difference between the rainfall accumulated so far this season (up to day t-1) and the historical mean for the same calendar day, divided by the historical standard deviation.
Formula	$$Z_{\text{accum}}(t-1) = \frac{\sum_{i \in \text{season},\ i < t} P_i - \mu_{\text{s2d}}(DOY_{t-1})}{\sigma_{\text{s2d}}(DOY_{t-1}) + \varepsilon}$$ mu_s2d and sigma_s2d are the mean and standard deviation of season-to-date cumulative rainfall at each calendar day, from the historical baseline.
Purpose	Quantifies whether the current year is pacing as a drought year (Z below -1.5), a normal year (Z between -0.5 and +0.5) or a surplus year (Z above +1.5).
Why it helps	Lets the model condition its seasonal quantile predictions on early-season performance.
Implementation note	Compute the baseline mean and standard deviation from the training years only. Using all 125 years would leak test-period information into the feature.